Homework 4: From PDEs to Kernels

UG+RPg **Presentations:** April 16 **Written report:** April 22
RPg only **3 Minute Thesis video submission:** April 22 **Screening:** April 23

All undergraduate and postgraduate students should attempt *all questions 1–3*. Each question is designed for guided self-study with the help of AI tools. Use AI to outline proofs, test ideas, or confirm derivations, but make sure your report clearly explains your own reasoning and interpretation.

Postgraduate (RPG) should present a short overview of their own in a short "3 Minute Thesis" video (about 3 minutes, max file size 100 MB) by April 22. We will watch selected videos together on April 23 as the final class session. (Use as much AI as you can)

1. **Existence and Uniqueness of PDE Solutions**
   Consider the boundary value problem, called the Poisson equation,

$$-\Delta u(x) = f(x), \quad x \in \Omega \subset \mathbb{R}^d, \qquad u|_{\partial\Omega} = g(x) = 0.$$

   Undergraduates may restrict to the one-dimensional case

$$-u''(x) = f(x), \quad x \in (0,1), \qquad u(0) = u(1) = 0.$$

   (a) Study how this equation can be formulated in a Sobolev space. Identify clearly:

   - which function space contains the *weak-form* solution u;
   - which class of test functions is used;
   - and what assumptions make the bilinear form *continuous* and *coercive*.

   (b) Learn the *Lax–Milgram theorem* and explain how it fits here. Outline the proof idea, then apply it to show existence and uniqueness of the Poisson problem.

   (c) *[RPG only]* Find out how additional regularity of f and a nonzero boundary function g can imply $u \in H^k$ for $k \geq 2$, referring to standard *elliptic regularity* results and explaining in what sense these improve the weak solution.

2. **Kernel Ridge Regression (KRR).**
   We observe n samples $(X_i, Y_i)_{i=1}^n$ generated by

$$Y_i = f_0(X_i) + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2),$$

   where the covariates X_i are i.i.d. samples from a probability measure μ supported on $\Omega = [0,1]^d$. Let k be a positive definite kernel on Ω, and let $\mathcal{H}_k$ denote the associated reproducing kernel Hilbert space (RKHS) with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}_k}$ and norm $\| \cdot \|_{\mathcal{H}_k}$. The RKHS provides the function space (trial space) for our estimator.

   (a) The KRR estimator is defined as the minimizer of the regularized least-squares functional

$$\hat{f}_{n,\lambda} \in \arg\min_{f \in \mathcal{H}_k} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}_k}^2 \right\}, \quad \lambda > 0.$$

   This optimization is posed in the infinite-dimensional space $\mathcal{H}_k$. Use the *representer theorem* to show that the minimizer can be expressed as a finite expansion

$$\hat{f}_{n,\lambda}(x) = \sum_{i=1}^n \alpha_i \, k(x, X_i),$$

and derive the corresponding linear system for the coefficients $\alpha = (\alpha_1, \ldots, \alpha_n)^\top$.

(b) We evaluate the estimator by its expected squared prediction error in $L^2(\mu)$ (the space of square integrable functions with respect to μ):

$$R_n(\hat{f}_{n,\lambda}, f_0) = \mathbb{E}_{f_0}^{(n)}\Big[L(\hat{f}_{n,\lambda}, f_0)\Big] \quad \text{with} \quad L(\hat{f}_{n,\lambda}, f_0) = \|\hat{f}_{n,\lambda} - f_0\|_{L^2(\mu)}^2.$$

Prove (or outline a proof) that, under suitable smoothness assumptions on f_0 and standard eigenvalue conditions on the kernel integral operator $T_k : L^2(\mu) \to L^2(\mu)$ defined by

$$[T_k g](\cdot) = \int_\Omega k(\cdot, x) g(x) d\mu(x),$$

the KRR estimator satisfies

$$R_n(\hat{f}_{n,\lambda}, f_0) \;\longrightarrow\; 0 \quad \text{as } n \to \infty,$$

that is, it converges to $f_0 \in \mathcal{H}_k$ in mean-square (expected L^2) sense. Discuss briefly what role the regularization parameter $\lambda = \lambda_n$ plays in ensuring this convergence.

3. Minimax convergence rate for KRR.

A *minimax convergence rate* describes the best possible worst-case rate at which an estimator approaches the true function over an entire model class. Let the data model, kernel, and RKHS setup be the same as in Question 2. Recall that $R_n(\hat{f}_{n,\lambda}, f_0)$ denotes the expected squared prediction loss for a fixed f_0.

(a) Explain carefully the meaning of the minimax risk

$$R_n^* = \inf_{\hat{f}_n} \sup_{f_0 \in \Theta} R_n(\hat{f}_n, f_0),$$

where $\Theta \subset \mathcal{H}_k$ is the model class of admissible true functions. Identify and interpret in words: *parameter space* Θ (what does Nature choose?), the *loss function* (what is evaluated?), the *infimum over estimators* (what can the statistician tune or optimise?), the *supremum over* f_0 (what does "worst-case over the model" mean?). Briefly discuss why R_n^* serves as a theoretical benchmark for any estimation method.

(b) Now restrict attention to the KRR estimator $\hat{f}_{n,\lambda}$. Derive (or outline) its mean-square error decomposition, separating the *bias* and *variance* contributions in $R_n(\hat{f}_{n,\lambda}, f_0)$. State clearly what quantities are random (depending on the noise and design) and what quantities are fixed (such as f_0, λ, and k). Explain how selecting $\lambda = \lambda_n$ balances these two terms as $n \to \infty$.

(c) *[RPG only]* Assuming $f_0 \in \Theta \subset \mathcal{H}_k$ (choose appropriate Θ) and the usual eigenvalue decay $\mu_j(T_k) \asymp j^{-2s/d}$ for the kernel integral operator T_k, state the asymptotic order of the minimax risk,

$$R_n^* \asymp n^{-2s/(2s+d)}.$$

Discuss briefly how the smoothness s, the dimension d, the noise level σ^2, and the kernel choice jointly determine the minimax optimal rate, and the optimal scaling of λ_n that allows KRR estimator to achieve this rate.

Report guideline: Questions 1–3 should be summarized in about 1–2 pages each, including the main theorem or result, a clear explanation or verification (steps may be supported by AI reasoning), and at least one short example, computation, or figure illustrating the concept.